TEMPLAT

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

– **TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2**

This document contains the following Sections

KEY PROJECT INFORMATION

GS ID of Project	11140
Title of Project	Improved biomass cookstoves for returnees in Burundi
Time of First Submission Date	16/03/2021
Date of Design Certification	
Version number of the PDD	5
Completion date of version	20/10/2022
Project Developer	OBEN EAC S.A
Project Representative	AERA Group SAS
Project Participants and any communities involved	AERA Group SAS OBEN EAC S.A
Host Country (ies)	Burundi
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	-
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 3.1 – Published August 2017
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	**SDG Impact (defined in**Error! Reference source not found.**)**	**Estimated Annual Average**	**Units or Products**
1 No poverty	Average households savings	26.46	$ in savings per year per household
7 Affordable Clean Energy	Number of beneficiaries	72,000	ICS units per year
13 Climate action	Amount of GHG Emission avoided	585,444	tCO2e per year
8 Decent Work and Economic growth	Total number of employees earning above local minimum wages	30	Number of jobs created with wages above minimum
15 Life on Land	Total non-renewable woodfuel saved.	303,403	t of wood saved/year Wood saved(ton)

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

According to the United Nations High Commission for Refugees (UNHCR), the UN refugee agency, 2019 year-end report[1], *"a persistent pattern of human rights violations which led over 400,000 Burundian nationals to flee since April 2015."* Furthermore, there were many internally displaced persons (IDPs) within Burundi too due mainly to natural disasters (floods) and climate change induced food insecurity. The UNHCR report also highlights that many refugees have started to return to Burundi (an excess of 21,000 in 2019) and after Burundi's involvement in the December 2019 Global Refugee Forum these numbers are expected to increase into the future.

The present project is a large-scale project that aims to provide returning refugees[2] from Burundi with a locally produced efficient or improved cookstove (ICS) as part of a package provided by aid agencies to returnees. The project will be located in Burundi.

[1] UNHCR: *Operation Burundi 2019 year-end report,* 07/07/2020

[2] UNHCR: *Glossary of terms: Definition of returnee, "A person who was of concern to UNHCR when outside his/her country of origin and who remains so, for a limited period (usually two years), after returning to the country of origin. The term also applies to internally displaced persons who return to their previous place of residence.",* https://reporting.unhcr.org/glossary/r accessed 6 January 2021

OBEN E.A.C is the project implementer and the manufacturer of the ICS.
The ICS distributed through this project will encourage returnees to replace the prevailing inefficient three-stone stoves or traditional cookstoves that are predominantly used across the region. In addition to the returnees, ICS may also be distributed to neighboring households of returnees in order to avoid any feeling of unfairness. The ICS provides for an efficient use of the thermal energy generated through the combustion of biomass while cooking and there is a more efficient transfer of this thermal energy to the cooking pot. There is thus a resultant reduction in both biomass consumption and greenhouse gas emissions.

1) Location of the project activity:

The project will target all the provinces of Burundi. The project is targeting returnees and neigbouring populations through all the Burundi. The targeted population primarily source of energy is woodfuel for cooking and the predominant technology is three-stone stoves or traditional cookstoves. The project implementer will capture data from each household equipped to ensure that household using woodfuel are targeted. If cookstoves are distributed to households using another type of fuel, then Emission Reductions will not be claimed for these households.

2) Technologies and measures to be employed and/or implemented by the project activity

2 types of cookstoves OBEN Matawi fixe and OBEN Matawi mobile will be equip beneficiaries' households. The project ICS combust biomass fuels more efficiently, reducing the greenhouse gas (GHG) emissions and particulate emissions (PM), thus improving the indoor air quality in project households. Due to the higher thermal efficiency of the ICS relative to the 3-stone fires, the ICS reduce the amount of non-renewable biomass fuel required for meeting similar thermal energy needs.

3) Project boundaries

The project boundary is Burundi, country where returnees are installed and where the fuel is collected.

4) Baseline scenario.

In the absence of the proposed project, inefficient three-stone stoves or traditional cookstoves fueled by firewood are used for cooking by returnees and their replacement with project ICS reduces non-renewable biomass fuel consumption, saving greenhouse gas emissions.

The project seeks to distribute up to 400,000 ICS (90,000 mobiles and 310,000 fixed) to returnees (targeted population), internally displaced person and neighbors. These ICS will be developed and distributed through official channels across all Burundi provinces. All the cookstoves will be distributed inside the geographical boundaries of Burundi;
In addition to the reduction of biomass consumption (and associated reduction in expenses to purchase wood – or reduction in time gathering firewood), there will also be an economic benefit in that the ICS will be produced locally in Bujumbura.

Leveraging carbon finance, this project will allow the reduction of greenhouse gas emissions for cooking and allow for the more sustainable use of biomass resources within Burundi.

A.1.1. Eligibility of the project under Gold Standard

The Gold Standard general eligibility criteria are documented in the Gold Standard for the Global Goals (GS4GG) Principle and requirements[3] specifically in section 3. According to Gold Standard for the Global Goals (GS4GG) Principle and requirements para 4.1.3: "A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it or it's referenced in the Gold Standard Product Requirements. There is a methodology associated with the activity: "Technologies and Practices to Displace Decentralized Thermal Energy Consumption V3.1." and the GS has published the Community Services Activity Requirements which include end-use energy efficiency projects, under which the project activity falls. Hence, the project activity falls under the automatic eligibility list of projects as the proposed project aims at distributing Improved Cook Stoves.
The project is located in Burundi. Burundi doesn't have an emission reduction cap enforced nor a emission trading system.

Eligibility criteria as per section 3.1.1 of GS4GG Principles & Requirements:

Eligibility Criteria	Justification
(a) Type of project: **Eligible Projects shall include physical action/implementation on the ground. Pre-identified eligible Project types are identified in the Eligibility Principles and Requirements section.**	The projects involve the distribution of Improved Cookstoves (ICS) . The project type is eligible under Community Services Activity Requirements .3.1.1(b)
(b) Location of project: **Projects may be located in any part of the world**	This project is located inside the boundaries of Burundi. See section A.2 below
(c) Project area, boundary and scale: Section 3.1.1 (c) : **The Project Area and Project Boundary shall be defined. […]and Products Requirements.** **In order to avoid double counting the […]of impacts amongst projects).**	The proposed project activity is a large-scale energy efficiency project activity, and the project area as well as the boundaries are clearly defined in A.1. and section A.2 The project takes place inside geographical boundaries of the republic of Burundi. The project achieves 585,444 tCO2 emission reduction per year on average and thus qualify as a large scale project This project is not included by any other carbon standard neither voluntary or compliance. To avoid inclusion of any stove which is a part of another registered carbon project/ programme, all ICS under this project shall have:

[3] Gold Standard: *GOLD STANDARD FOR THE GLOBAL GOALS PRINCIPLES & REQUIREMENTS Version 1.2,* October 2019

	• a unique ID serial number • End User data (name/address/physical location/phone number/govt. ID number) • Records of baseline stove type and fuel type used. This data will uniquely identify the ICS, avoiding any double counting and trace its user for future monitoring and verification. There is quite no chance that the targeted population are already involved in another voluntary or compulsory programme, as most of them are returnees, that are coming back after fleeing to neighboring countries. OBEN E.A.C is the only company involved in the distribution of cookstoves as part of the aid package. However, prior to distribution, it will be check that households are not already engaged in another carbon finance project. Households with improved cookstoves will not be equipped with this project ICS.
(d) Host country requirements : **Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.**	This project is in compliance with Burundi's legal, environmental, and social regulations. The project is actively contributing to the Host Country by reducing the GHG emissions, fuel consumption as well as economic and time savings to thousands of poor households
(e) Contact Details: **As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details […]to refuse use of the Standard where reputational concerns are highlighted**	All the project participant informations have been provided in Appendix 2
(f) Legal ownership **Full and uncontested legal ownership[…]All projects shall immediately report to Gold Standard any land title/tenure disputes arising.**	For each distribution, a Carbon Transfer Form (CTF) will be signed and uploaded to our database stating that the rights to the carbon credits will lie with the project developper.which is OBEN E.A.C
(g) Other Rights : **As well as legal title and ownership[…] Project implementation in affected areas**	There are no disputes or contested rights that have been identified in relation to rights relevant to the project activity

(h) Official Development Assistance (ODA) Declaration: All Project Developers applying for project activities […] shall declare the Official Development Assistance (ODA) support.	It is confirmed that no Official Development Assistance (ODA) has been requested and none will be applied for the project activity.

Eligibility criteria as per GS4GG Community Services Activities Requirements

Eligibility Criteria	**Justification**
2) Eligibility project types	
2.1.2) All CSA projects shall lead to climate change mitigation and/or adaption by providing or improving access to services/resources at household or community level or institutional level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	By providing improved cooking stoves to communities, the project improves access to energy services/resources at community level. As such, the project is an Eligible Project Type in line with the requirements
2.1.3) Projects shall conform to the Principles and Requirements	The project conforms with the Principles and Requirements detailed in the document. The project is eligible under section 4, Principle 1, section (a) of the Principles and Requirements as it follows an established Gold Standard methodology TPDDTEC V3.1. Concerning point 4.1.7, the project does not support geoengineering or entail energy production from fossil fuels or nuclear.
3 General Eligibility Criteria	
3.1.1 Types of Project Type of project: Section 3.1.1 (b) identifies “End-use energy efficiency” projects as eligible and uses an example of “efficient cooking”.	This project activity meets this criterion as it is an efficient cookstove project activity. The project aims at distributing ICS to households. ICS have a higher thermal efficiency and then falls in the efficient cooking category
3.1.2 Project Area, boundary and scale Project area, boundary and scale: Section 3.1.2 (b) :Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements	The proposed project activity is a large scale energy efficiency project activity, and the project area as well as the boundaries are clearly defined in A.1. and section A.2 The project takes place inside geographical boundaries of the republic of Burundi. The project achieve 585,444 tCO2 emission reduction on average per year and thus qualify as a large scale project
3.1.4 Legal ownership a) Projects involving the distribution of a large number of devices for services shall provide a clear description of the ownership of the	a) The ownership is defined in section A.1.2 below. The stove recipients are clearly explained by the local partner about the transfer of the ownership rights and selling the emission

Products that are generated under Gold Standard Certification all along the investment chain. In line with FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. b) The transfer for Product ownership shall be discussed during the local stakeholder consultations for projects.	reductions resulting from the project activity once they accept the stove. They are explained that by signing the Carbon Transfer Form and providing their contact information for monitoring purposes, they enter into agreement that the VERs are transferred to the project developer bThe tranfer of ownership have been discussed during the local stakeholder consultations. During both the physical meeting and remote consultation, participants have been informed about the transfer of ownership as also mentioned in the end-users' agreements

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

OBEN EAC S.A the project owner (Observatoire Burundais de l'Environnement et de la Nature, a local company) has granted full and uncontested legal ownership of the emission reductions that are generated under this Gold Standard project to Aera Group SAS, and has legal rights concerning changes in use of resources required to service the Project for ownership of emission reductions.

A.2. Location of project

The project activity will be located solely within the geographic borders of the Republic of Burundi in central Africa. The project activity will potentially occur within all regions and communities of Burundi.

Figure 1: Map of the project boundary

A.3. Technologies and/or measures

As specified in section A.1, 2 types of ICS (OBEN Matawi fixe and OBEN Matawi mobile) are being distributed under this project. The ICS distributed under the project are manufactured following the" Jiko matawi stove style, a Tanzanian-design ICS that is produced from moulded clay which is dried through high temperatures in a kiln to reinforce its properties including high thermo efficiency and durability, or alternatively dried in a cool, dry, ventilated area out of direct sunlight to cure the final products. The stoves distributed are simple to use, and are multi-purpose stove using both wood and charcoal. The 2 types of the stoves being installed are:

- A mobile model that is made of clay, and reinforced with a metal protection and a
- Fixe model same as the mobile model but now reinforced by a brick structure and fixed inside kitchen for more permanent use.

Figure 2: Fixed Matawi

Figure 3 : Mobile Matawi

Weight	5.1 kg (stand-alone)
Efficiency	27.4% (WBTs results)[4]
Type of fuel	Wood/Charcoal
Firepower	5,4-5,8 kW
Height x Width	23 x 26 cm
Recommended pots diameter	12 inches
Durability	Approx. 8 years

Figure 3:Mobile ICS specifications

[4] Based on average results from WBTs (Mobile model WBTs report) conducted for the project

Weight	5.1 kg (stand-alone)
Efficiency	31.4%[5] (new)
Type of fuel	Wood/Charcoal
Height x Width	23 x 26 cm
Firepower	5,4-5,8 kW
Recommended pots diameter	12 inches
Durability	Approx. 8 years

Figure 4 :Fixed ICS specifications

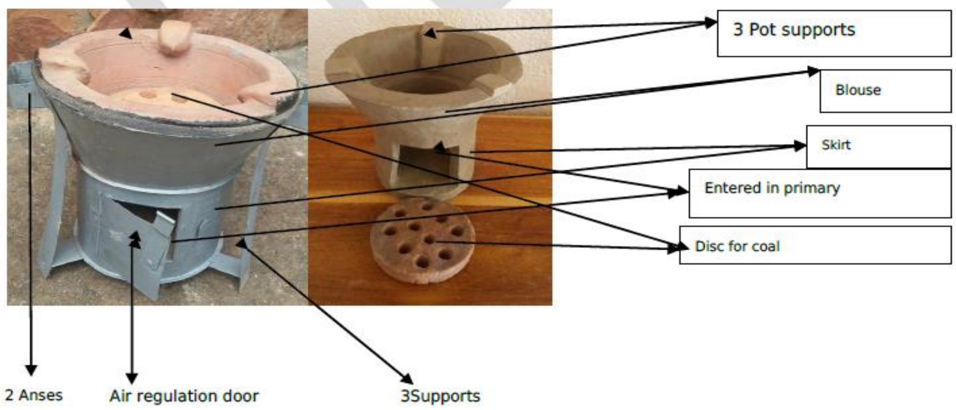

Figure 6: ICS design and component parts

A.4. Scale of the project

The project is a large scale project activity as per GS community services activity requirements V1.2

A.5. Funding sources of project

Project financing for this project activity will not use Official Development Assistance (ODA) Funds. The project developer has signed the ODA declaration template.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

Gold Standard Methodology - Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 3.1, August 2017

[5] Based on average results from WBTs (Fixed model WBTs report)

Project type: (b) End - Use Energy efficiency (Gold standard Community Service Activity Requirements V 1.2 para 3.1.1 (b))

B.2. Applicability of methodology (ies)

The justification for the choice of methodology is presented in the table below.

Applicability Criteria	Justification
§1.0 1 - The project boundary needs to be clearly identified, and the technologies counted in the project are not included in any other voluntary market or CDM project activity (i.e., no double counting takes place). In some cases, there may be another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	The project boundary can be and is clearly defined in section B.3 below. The ICS distributed as part of this project activity will have a project identifier and unique serial number inscribed in the metal. This will prevent any other project proponent, organization, retailer, individual or distributors from using the same ICS and including it in another project activity. The ICS included in the project activity are not part of any other voluntary market, CDM project activity or other carbon finance project. Checks in place will confirm that there is no double counting and these will include, inter alia, records of serial numbers and returnees receiving each ICS; interviews with ICS owners and users; monitoring surveys by independent consultants and/or activities of the VVB during their verification phases.
§ 1.0 2 – The technologies each have continuous useful energy outputs of less than 150kW per unit (defined as the total useful energy delivered from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	Independent WBTs show that both types of OBEN ICS have useful energy outputs of less than 150kW. The results showed useful energy between 5.493 – 5.823 kW_{th} (WBTs reports). This is less than the stipulated 150 kW useful energy threshold of the methodology.
§ 1.0 3 – Using the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology (e.g., discounted price for the improved technology) and the definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for	As the ICS owners will be returnees returning from either IDP camps and locations or relocating from other countries, they are unlikely to have significant possessions, but there will be no program (apart from awareness and sensitization) to prevent the ICS owners from using the older technologies. The fact that the OBEN ICS will be provided free of charge as part of a returnee aid package, there is an economic encouragement for the returnees to not purchase another stove for cooking that may be less efficient. Any older technology stoves that may be in pace (along with the

a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology. For example, whether the existing baseline technology is not surrendered at the time of the introduction of the improved technology, or whether a new baseline technology is acquired and put to use by targeted end users during the project crediting period. The success of the mechanism put into place must therefore be monitored, and the approach must be adjusted if proven unsuccessful. If an old technology remains in use in parallel with the improved technology, the corresponding emissions must be accounted for as part of the project emissions	inefficient and traditional three stone cooking fire) will be available as a backup or auxiliary technology for the ICS owners. Over time, the positive benefits of the stoves (reduction in fuel requirements, reduction in indoor smoke and associated health benefits etc.) will act as an incentive for users to discontinue use of any older less efficient cooking stoves.
§ 1.0 4 – The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. For technology producers and the retailers of the improved technology or the renewable fuel in use, this must be communicated by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end users, the end users will need to be informed and notified that they cannot claim for emission reductions from the project.	Each ICS end user, who is the default owner of emission reductions, is notified that they waive ownership of ERs upon receipt of each stove in their package given from the aid agencies. This is done via a right waiver that is included with each stove when distributed to make the customer aware of them waiving ownership rights over emission reductions. Each stove owner (returnee) is given their ICS as part of a larger aid package and they do not purchase the ICS specifically.
§ 1.0 5 – Project activities making use of a new biomass feedstock in the project situation (e.g., shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules. If the biomass feedstock is sourced from a dedicated plantation, the criteria must apply to both plantations established for the project activity AND existing plantations that were established in the context of other activities but will supply biomass feedstock.	The project activity is not making use and will not make use of any new biomass feedstock in the project situation.
Furthermore, the following conditions apply:	
§ 1.0 5 (a) – Adequate evidence is supplied to demonstrate that indoor air pollution (IAP) levels are not worsened compared to	The project activity is not making use and will not make use of any new biomass feedstock in the project situation.

the baseline, and greenhouse gases emitted by the project fuel/stove combination are estimated with adequate precision. The project fuel/stove combination may include instances in which the project stove is a baseline stove.	
§ 1.0 5 (b) – Records of renewable fuel sales may not be used as sole parameters for emission reduction calculation, but may be used as data informing the equations in section 2.0 of this methodology. These records need to be correlated to data on distribution and results of field tests and surveys confirming **(a) actual use of the renewable fuel and usage patterns (such as average fraction of non-renewable fuels used in mixed combustion or seasonal variation of fuel types),** **(b) GHG emissions,** **(c) evidence of CO levels not deteriorating** **(d) any further factors effecting emission reductions significantly.**	The project activity is not making use and will not make use of any new biomass feedstock in the project situation.

B.3. Project boundary

The project activity involves dissemination of ICS to returnees as part of an aid package, cutting consumption of non-renewable biomass and corresponding greenhouse gas emissions in Burundi. The geographical project boundary is the geographic borders of the Republic for Burundi. In the baseline and project scenario, the traditional and improved cook stoves use firewood that is sourced from farmlands, grasslands and forests within Burundi.

The project boundary is specified in the methodology as follows:
a. The project boundary is the physical, geographical sites of the project technologies. This boundary could also host the baseline and project fuel collection and production (e.g., charcoal, plant oil) and solid waste and effluents disposal or treatment facilities associated with fuel processing.

The project boundary is therefore defined as the domestic kitchens in which each project technology is installed.

b. The target area is the region(s) or town(s) where the considered baseline scenario(s) are deemed to be uniform across political borders. This area could be within a single country, or across multiple adjacent countries. The target area provides an outer limit to the project boundary in which the project has a target population.

The target area is defined in Section A.2 above, which corresponds to the geographical borders of the Republic of Burundi.

c. In cases where woody biomass (including charcoal) is the baseline fuel or where the project activity introduces the use of a new biomass feedstock into the project situation, the fuel production and collection area is the area within which this woody or new biomass is produced, collected and supplied.

The fuel production and collection area is within the target area defined above in A.2, which corresponds to the geographical borders of the Republic of Burundi.

	Source	GHGs	Included?	Justification/Explanation
Baseline	Combustion of woodfuel for cooking	CO2	Yes	Significant source of emissions
		CH4	Yes	Significant source of emissions
		N2O	Yes	Significant source of emissions
Project	Combustion of woodfuel for cooking	CO2	Yes	Significant source of emissions
		CH4	Yes	Significant source of emissions
		N2O	Yes	Significant source of emissions

B.4. Establishment and description of baseline scenario

According to the applied GS methodology, a baseline scenario is defined by the typical baseline fuel consumption patterns in a population that is targeted for adopting the new project technology.
The majority of households in Burundi still rely on woody biomass for cooking. Most households use the three-stone-fire as their principal method of cooking. The baseline scenario is the usage by target population of firewood in traditional low efficiency firewood stove or three-stone fire for cooking The baseline scenario is identified as the use of firewood from non-renewable sources in traditional three stone cooking stoves or inefficient stoves.
The UN WFP[6] has identified that around 96% of energy requirements are from traditional biomass, "with inefficient use of firewood makes wood fuel increasingly scarce and places reforestation efforts in competition for land with agricultural production."
The baseline scenario has been established in line with GS4GG Principles & Requirements and the TPDDTEC V3.1 methodology. The baseline study consisted of 111 households using firewood, surveyed by the project implementer. The households were surveyed across 9 of the 18 provinces of Burundi (see table and map below).

Province	Number of respondant
BUBANZA	15
BURURI	19
CIBITOKE	19
GITEGA	10

[6] US AID – United States Agency for International Development: "USAID/Burundi Gender analysis. Final Report 2017", https://banyanglobal.com/wp-content/uploads/2017/07/USAID-Burundi-Gender-Analysis-Final-Report-2017.pdf, accessed 19 January 2021

MAKAMBA	10
MURAMVYA	10
MUYINGA	8
RUTANA	11
RUYIGI	9
Total	**111**

Table 1: Number of people surveyed by region

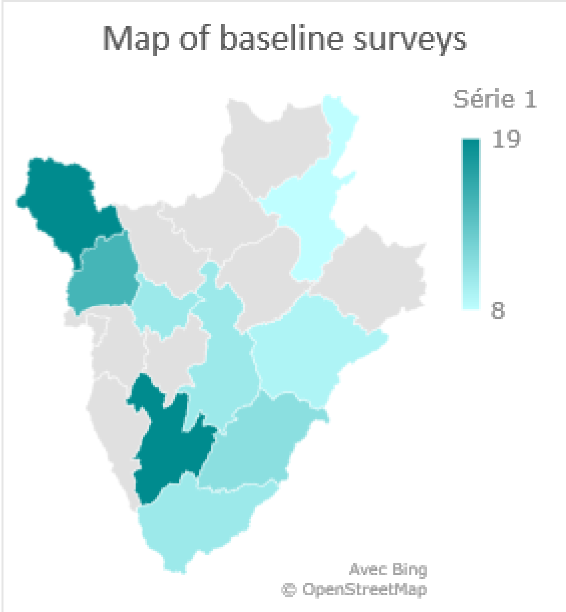

Figure 7: Map of people surveyed

The baseline study was done on population using firewood for cooking (baseline scenario). The results of the baseline showed an average household size of 8.9 persons. Taking a "Standard adult" equivalence factor, this leads to an average of 6.2 persons per households.
The average woodfuel consumption per day was equal to 10.1 kg per household. And all the household reported using three stone as main cooking technology.
On the sample surveyed, in 2.7% of the households (3), the man was in charge of fuel collection, while in other cases this duty was done by women (the wife or the daughter).
Regarding woodfuel acquision, 36% (40 households) off the household declared that they were purchasing woodfuel on the market, while 10% (11 households) were collecting firewood around their place and finally 54% (60 households) were both buying and collecting woodfuel. The household reported cooking on average 1.9 time per day.

The mains results are provided below:

Average Number of persons per households	8.9
Average Number of persons per households (standar adult equivalent)	6.2
Average woodfuel consumption per household (kg/year)	10.1
Fraction of household purchasing woodfuel	36%
Fraction of household collecting woodfuel	10%
Fraction of household purchasing and collecting woodfuel	54%

Gender and age	**Average number of people**	**Fraction of Standard adult**	**Adult equivalent**	**Average daily consumption of woodfuel (kg)**
Child : 0- 14 years	4,2	0,5	2,1	10.1

Female: over 14 years	2,8	0,8	2,2	
Male: 15-59 years	1,7	1	1,7	
Male: over 59 years	0,2	0,8	0,2	
Total	**8,9**		**6,2**	

Following the baseline surveys, a kitchen performance test (KPT) to determine the baseline woody biomass consumption was conducted between January and March 2022 on a paired sample of 70 households across the 7 Provinces within Burundi (project boundary) included the project boundary. 10 households were selected by province in each of the 7 provinces. The KPT was conducted by an independent expert, the Centre de Recherche Universitaire sur les Energies Alternatives (CRUEA). The Project KPT was conducted on households equipped with mobiles ICS. The 70 households were selected in the 111 households surveys, and the KPT, was conducted on the households before and after the introduction of ICS.
The KPT have been conducted during 3 consecutive days from Tuesday to Friday for each phase (before and after the ICS introduction).
The KPTs was conducted to assess the quantity of fuel saved by replacing baseline stoves (3 stones) with the mobile model of ICS. As the mobile model is less efficient than the fixed one, the results can be used as more conservative.
The KPTs resulted in an average woodfuel consumption of 11.08 kg/day/household (before the cookstove introduction, and an average woodfuel consumption of 5.5kg/day/household, hence a reduction of 5.58 kg/day/household (2,040 kg saved per household/year, rounded to 2 tonnes/household/year).

The value of the biomass consumption before the project (11.1 kg/hh/year or 4.05t/hh/year) is found conservative compared to available historical data such as the expired standardized baseline for Burundi[7] that refers to 1.1 t/p/year in rural areas and 1.5t/p/year in Urban areas. Thus, a consumption of 9.79 t/hh/year in rural areas and 13.35t/hh/year in urban areas.

The detailed household responses and the baseline survey/KPT analysis are provided in the Ex-ante calculation sheet and in the KPT report. The analysis showed that the results respected the 90/30 rule for paired sample.

Baseline 90/30 check	Before outliers removal (before introduction of ICS)	After outliers removal (before introduction of ICS)	Before outliers removal (after introduction of ICS)	After outliers removal (after introduction of ICS)
Mean	11,08	11,08	5,42	5,50

[7] https://cdm.unfccc.int/methodologies/standard_base/2015/sb65.html

Standard Deviation	2,10	2,10	0,83	0,67
COV	0,19	0,19	0,15	0,12
Upper Quartile	12,56	12,56	6,02	6,08
Lower Quartile	9,96	9,96	4,95	5,02
Interquartile Range	2,60	2,60	1,07	1,06
Outlier Threshold (Upper)	16,46	16,46	7,62	7,68
Outlier Threshold (Lower)	6,06	6,06	3,34	3,43
Valid sample		70		68
Standard error		0,25		0,08
What is the precision attained?		0,04		0,02
Does the result satisfy the 90/30 precision rule?	YES		YES	
What baseline value may be applied for calculation?	Use mean value		Use mean value	

Average fuel savings (kg/day/hh)	**5,58**
Average fuel savings (tonnes/year/hh)	**2,04(rounded to 2)**

Table 2: KPT results

PROVINCE	Number of surveys
Rutana	10
Gitega	10
Muramvya	10
Muyinga	10
Ruyigi	10
Bururi	10
Bubanza	10

Table 3: Regions covered by KPT

Although ICS may be distributed to households using charcoal, only the firewood users will be accounted for emissions reduction calculation. Charcoal users will be removed from the database

B.5. Demonstration of additionality

B.5.1. Prior Consideration

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	As per GS4GG Community services activity requirements, Version 1.2, Para 4.1.9, Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of design certification: (a) Positive list (Annex B of this document) (b) Projects located in LDC, SIDS, LLDC (c) Microscale projects As per Annex B – Positive list, Para 1.1.3 of GS4GG Community services activity requirements, Version 1.2, Project activities solely composed of isolated units where the users of the technology/ measure are households or communities or institutions and where each unit results in <= 1.8 GWhth of energy savings per year or <=600 tonnes of emission reductions per year.
Describe how the proposed project meets the criteria for deemed additionality.	The project distributes isolated units where the end-users are households. Each isolated unit(stove) result in less than 1,800 MWhth. Moreover, the project takes place in a LDC country and therefore is deemed additional.

In terms of §4.1.49 (b) of the GS4GG Principles and Requirement (v1.2), retroactive shall submit the required documents for preliminary review within one year of the project start date. This has been achieved as the project start date is identified in section C.1.1 below and the documents were submitted for preliminary within a one-year period.

B.5.2. Ongoing Financial Need

>>.N/A

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
1 No poverty	By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance	Distribution of ICS, increase access to clean cooking while reducing poverty by reducing purchased fuel consumption Indicator : Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking
7 Affordable and Clean Energy	7.1 By 2030, ensure universal access to affordable, reliable and modern energy services	This project involves dissemination of clean, modern technology for cooking, by using available energy sources more efficiently. Indicator : Number of beneficiaries : Households
13 Climate Action	13.2 Integrate climate change measures into national policies, strategies and planning	This project is directly related to the target as it contributes towards avoidance of GHG emissions from replaced of fossil fuel / non-renewable biomass-based cooking.

		Indicator : Amount of GHGs emissions avoided or sequestred
15 Life on land	15.1: By 2020, **ensure the** conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in line with obligations under international agreements	Indicator : Total non-renewable wood fuel saved
8 Decent Work for All	8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Indicator : Total number of employees earning above local minimum wage

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

SDG1: No poverty

Applied methodology/approach	Equation/calculation
Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking Approach: Calculation of household savings via quantity of wood fuel saved by stove usage and average fuel prices.	Ex-post Monitoring Survey Records measuring money savings due to reduced fuel consumption in households MS = By * Up,y * Price of fuel Where: MS — Money savings per year per household By — Quantity of firewood that is saved in the year y (tonnes per ICS in year y) Up,y — Usage rate in project scenario p during year y

SDG 7 Affordable and Clean Energy

Applied methodology/approach	Equation/calculation
7.1.2 Proportion of population with primary reliance on clean fuels and technology Approach: Monitor the number of ICS distributed under the project as an indicator of providing clean technology (relative to baseline stoves).	ICS distribution records ACS Net Benefit (SDG 7) = Number of beneficiaries * Usage rate of the cookstoves

SDG 8 Decent work

Applied methodology/approach	Equation/calculation

8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities Approach: Monitor the total number of employeesearning above the minimun monthly wages in Burundi.	Number of decent jobs created (NDJ) (Total number of employees earning above local minimum wage).

SDG 15 Life on Land

Applied methodology/approach	Equation/calculation
15.1.1 Forest area as a proportion of total land area Approach: Monitor the quantity of wood saved by year	ICS distribution records Net Benefit (SDG 15) WS = By * Number of ICS operational*fNRB Where: WS Tons of wood saved By Quantity of firewood that is saved in the year y (tonnes per ICS in year y)

SDG 13 Climate Action

Applied methodology/approach	Equation/calculation
13.2.1 Amount of CO_2e emissions reduced by the project per year Approach: TPDDTEC, version 3.1	A fixed non-renewable biomass fraction value, determined and fixed ex-ante, as prescribed in the methodology, will be established for the given host country. Baseline surveys should be carried out as per the sample sizing mentioned on page 12 of the methodology. The baseline surveys shall collect information on the following (but not limited to): 1. End user details 2. Number of people served 3. Type of baseline devices and service level (Commercial / domestic) 4. Types of fuels used and quantities 5. Estimated frequency (number of meals cooked) 6. Seasonal variations etc. The baseline fuel consumption, established via in-field KPTs for a given host country, may be fixed for the entire crediting period. For ex-post emission reduction calculations, baseline KPTs will be conducted by PP before requesting first verification. For subsequent verifications the PP may conduct baseline KPTs biennially, if

	they don't wish to use the baseline KPT value established before first issuance. Otherwise, a default value of 0.5 tonnes per capita per year in case of baseline fuel is firewood, may be used. In this case, the number of people served should be monitored to determine total baseline fuel consumption per sampled entity. Project non-renewable biomass assessment may be deemed same as baseline, albeit should be updated for project fuel mix. Project surveys should be carried out as per the sample sizing mentioned on page 12 of the methodology. The project surveys shall collect information on the following (but not limited to): 1. End user details 2. Number of people served 3. Type of project devices and service level (Commercial / domestic) 4. Types of fuels used and quantities 5. Estimated frequency (number of meals cooked) 6. Seasonal variations etc. The project fuel consumption, established via in-field KPTs for a given host country, will be conducted at-least biennially. The baseline and project KPTs will be determined using either paired / independent or single sample tests. 90/30 rule for sample size determination will be applied in case of paired or independent sampling. 90/10 rule will be applied in case of single sampling. In case desired precision is not achieved, lower bound value of the 90% confidence interval should be applied. The overall GHG reductions achieved by the project activity will be calculated as follows: $ER_y = \sum_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO2} + EF_{fuel,nonCO2})) - \sum LE_{p,y}$ Where: $\sum_{b,p}$ Sum over all relevant (baseline b/project p) couples

	$N_{p,y}$ Cumulative number of project technology-days included in the sales/distribution database for project scenario p against baseline scenario b in year y $U_{p,y}$ Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage surveys (fraction). A separate usage factor is determined for each technology in the project. $P_{p,b,y}$ Specific fuel savings for an individual technology of project p against an individual technology of baseline b in year y, in tons/day, as derived from the statistical analysis of the data collected from the field tests $NCVb_{,fuel}$ Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton) fNRB,b,y Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass (drop this term from the equation when using a fossil fuel baseline scenario) EFfuel,CO2 CO2 emission factor of the fuel that is substituted or reduced. 112 tCO2/TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel $EF_{fuel,nonCO2}$ Non-CO2 emission factor of the fuel that is reduced $LE_{p,y}$ Leakage for project scenario p in year y (tCO_2e/yr) Leakage, if applicable, will be assessed on the following points:

	a. The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project. b. The NRB or fossil fuels saved under the project activity are used by non-project users who previously used lower emitting energy sources. c. The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario. d. The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology. e. By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.

B.6.2 Data and parameters fixed ex ante

SDG13

Data/parameter	$EF_{b,i,CO2}$
Unit	tCO2/TJ
Description	CO_2 emission factor arising from use of fuel type i in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5)
Value(s) applied	112

Choice of data or Measurement methods and procedures	Default IPCC value for fuel wood is applied.
Purpose of data	For emission reduction calculation
Additional comment	

Data/parameter	$EF_{p,i,CO2}$
Unit	tCO2/TJ
Description	CO_2 emission factor arising from use of fuel type i in project scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5)
Value(s) applied	112
Choice of data or Measurement methods and procedures	Default IPCC value for fuel wood is applied.
Purpose of data	For emission reduction calculation
Additional comment	

Data/parameter	$EF_{b,i,non,CO2}$
Unit	tCO_2/TJ
Description	Non-CO_2 emission factor arising from use of fuels in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5); IPPC AR5 values for Global Warming Potential of CH4 and N_2O
Value(s) applied	9.46
Choice of data or Measurement methods and procedures	Data from Table 2.5, volume 2, chapter 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
Purpose of data	For emission reduction calculation
Additional comment	Data from Table 2.5, volume 2, chapter 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Emission factor for CH4= 300kg CH4/TJ Emission factor for N_2O= 4kg N20/TJ Source: *IPCC* Global Warming Potential (AR5)

	GWP CH4= 28 GWP N_2O = 265 Source :*IPCC AR5 (GS4GG)* EFb,i,non,CO2 = 300*28 + 4*265 = 8,400+1,060= 9,460 kgCO2e/TJ= 9.46 tCO2/TJ

Data/parameter	$EF_{p,i,non,CO2}$
Unit	tCO_2/TJ
Description	Non-CO_2 emission factor arising from use of fuels in project scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, volume 2, chapter 2 (Table 2.5); IPPC AR5 values for Global Warming Potential of CH4 and N_2O
Value(s) applied	9.46
Choice of data or Measurement methods and procedures	IPCC default value. Deemed valid by the methodology
Purpose of data	For emission reduction calculation
Additional comment	Data from Table 2.5, volume 2, chapter 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Emission factor for CH4= 300kg CH4/TJ Emission factor for N20 = 4kg N20/TJ Source: *IPCC* Global Warming Potential (AR5) GWP CH4= 28 GWP N20 = 265 Source :*IPCC AR5 (GS4GG)* EFb,i,non,CO2 = 300*28 + 4*265 = 8,400+1,060= 9,460 kgCO2e/TJ= 9.46 tCO2/TJ

Data/parameter	$NCV_{b,i}$
Unit	TJ/tonne
Description	Net calorific value of the fuel type *i* used in the baseline
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 1: Introduction, Table 1.2 - Default net calorific values
Value(s) applied	0.0156

Choice of data or Measurement methods and procedures	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 1: Introduction, Table 1.2 - Default net calorific values
Purpose of data	For emission reduction calculations
Additional comment	

Data/parameter	$NCVp_{,i}$
Unit	TJ/tonne
Description	Net calorific value of the fuel type *i* used in the project scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 1: Introduction, Table 1.2 - Default net calorific values
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	Default IPCC values have been applied for woodfuel
Purpose of data	For emission reduction calculations
Additional comment	

B.6.3 Ex-ante estimation of SDG impacts

SDG 1: No Poverty

Net benefit (SDG 1) = MS = By * Up,y * Price of wood fuel = 26.46

Where:

MS Money savings in USD $ per household per year

Up,y Usage rate in project scenario p during year y = 0.9 (TPDDTEC Cookstoves usage rate guidelines good practices)

By Quantity of fire wood that is saved in the year y (tonnes per ICS in year y) = 2 tonnes (KPT report)

Price of wood fuel is estimated at 0.0147 $ per kg[8] in Burundi.

SDG 7: Affordable and Clean Energy

ICS distribution records

Net Benefit (SDG 7) = Number of beneficiaries * Usage rate of the cookstoves
= 400,000 (over the crediting period)*0.9 =360,000= 72,000 beneficiaries /year

SDG 8 : Decent job created

OBEN E.A.C Human resources management data

Net benefit (SDG 8) = $NDJ_{Project}$ - $NDJ_{Baseline}$ = 30 (ex-ante)

Where:

$NDJ_{Baseline}$ Number of employees earning above minimum wages under Baseline = 0
$NDJ_{Project}$ Number of employees under Project earning above minimum wages = 30 (ex-ante)

SDG 13: Climate Action

As per the methodology the governing equation for the emission reduction calculations is as follows[9] with ($\sum BE_{b,y} - \sum PE_{p,y}$) is directly merged in to the following equation.

$$ER_y = \sum_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO2} + EF_{fuel,nonCO2})) - \sum LE_{p,y}$$

Where:

Parameter		Value			
$\sum_{b,p}$	Sum over all relevant (baseline b/project p) couples	Year	Mobile ICS	Fixed ICS	Total
		2020	13,627	0	13,627
		2021	28,439	53,027	81,466
		2022	40,000	133,027	173,027
		2023	60,000	215,000	275,000
		2024	90,000	310,000	400,000
		2025	90,000	310,000	400,000
$N_{p,y}$	Cumulative number of project technology-days included in the sales/distribution database for project scenario p against	Year	Mobile ICS	Fixed ICS	Total
		2020	3,856,441	0	3,856,441
		2021	10,380,235	19,354,855	29,735,090
		2022	14,600,000	48,554,855	63,154,855
		2023	21,900,000	78,475,000	100,375,000

[8] Based on data from Institut de Statistiques et d'Etudes économiques du Burundi (May 2022), in March 2022, 1 stere of wood fuel was sold at 12,094 FBU (complete calculation for 1 kg of wood fuel available in SDG Impact Tool)

[9] The ERs are being determined based on equation (1) of the methodology, page 19.

<table>
<tr><td></td><td>baseline scenario b in year y</td><td><table><tr><td>2024</td><td>32,940,000</td><td>113,460,000</td><td>146,400,000</td></tr><tr><td>2025</td><td>7,380,000</td><td>25,420,000</td><td>32,800,000</td></tr></table></td></tr>
<tr><td>$U_{p,y}$</td><td>Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage surveys (fraction). A separate usage factor is determined for each technology in the project</td><td>90%</td></tr>
<tr><td>$P_{p,b,y}$</td><td>Specific fuel savings for an individual technology of project p against an individual technology of baseline b in year y, in tons/day, as derived from the statistical analysis of the data collected from the field tests</td><td>2t of wood/year (KPT results)</td></tr>
<tr><td>$NCVb_{,fuel}$</td><td>Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/ton)</td><td>0.0156 TJ/ton</td></tr>
<tr><td>fNRB,b,y</td><td>Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass (drop this term from the equation when using a fossil fuel baseline scenario)</td><td>81.83 %</td></tr>
<tr><td>EFfuel,CO2</td><td>CO2 emission factor of the fuel that is substituted or reduced. 112 tCO2/TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel</td><td>112 tCO2e/TJ</td></tr>
</table>

<table>
<tr><td>$EF_{fuel,nonCO2}$</td><td>Non-CO2 emission factor of the fuel that is reduced</td><td>9.46 tCO2e/TJ</td></tr>
<tr><td>$LE_{p,y}$</td><td>Leakage for project scenario p in year y (tCO_2e/yr)</td><td>0% (see below)</td></tr>
<tr><td>ER</td><td>Emissions reduction(tCO2e)</td><td>
<table>
<tr><td>Year</td><td>Mobile ICS</td><td>Fixed ICS</td><td>Total</td></tr>
<tr><td>2020</td><td>29,997</td><td>0</td><td>29,997</td></tr>
<tr><td>2021</td><td>80,743</td><td>150,552</td><td>231,295</td></tr>
<tr><td>2022</td><td>113,566</td><td>377,685</td><td>491,251</td></tr>
<tr><td>2023</td><td>170,349</td><td>610,419</td><td>780,768</td></tr>
<tr><td>2024</td><td>256,224</td><td>882,550</td><td>1,138,774</td></tr>
<tr><td>2025</td><td>57,405</td><td>197,730</td><td>255,135</td></tr>
</table>
</td></tr>
</table>

Leakage emissions assessment.

Potential leakage source	**Probability**	**Justification**
The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.	Can be excluded	All the stoves replaced are mostly inefficient firewood traditional stoves i.e three stone stoves that have no market value, and use is widespread. Therefore, it is highly unlikely that three stone stoves will be used outside the project boundary in replacement of lower emitting technology.
Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.	Can be excluded	Project users have to spend money or time for the firewood. It can be excluded that the fuel saved by the project would be given for free by the project users and used by non-project users who previously used lower emitting energy sources.
The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.	Can be excluded	The project is not big enough so that it would significantly have an impact on the NRB fraction. Besides, demand for firewood in Burundi is continuously rising. Since alternative fuels (like LPG or electricity) are out of reach for many people. Hence, the share of NRB remains high and it will not have a leakage impact on other carbon projects in Burundi.
The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.	Can be excluded	The mean temperature conditions in Burundi is 24.05° (https://www.climatestotravel.com/climate/burundi) and do usually not require space or room heating. It is also very unlikely that the baseline cookstoves will remain used for space heating. However, the continued usage will be monitored during annual surveys.

By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.	Can be excluded	The project's target group is households using firewood with ineficient traditional stoves as evidenced by the baseline report and the KPT. It is highly unlikely that households using LPG or electricity for cooking would use the project technology. Thus, leakage can be excluded.

Therefore leakages assessment is assessed to be zero.

SDG 15 : Life on land
ICS distribution record
WS (SDG 15) = Number of beneficiaries * Usage rate of the cookstoves* Bysavings* fNRB = 1,517,013 = 303,403 /year
Where:
WSJ Total non renewable wood (tons) saved under project scenario= 303,403(ex-ante)

B.6.4 Summary of ex ante estimates of each SDG outcome

SDG 1: No Poverty

Year	Baseline estimate	Project estimate	Net benefit
24 March 2020- 31 December 2020	0	19.85	19.85
2021	0	26.46	26.46
2022	0	26.46	26.46
2023	0	26.46	26.46
2024	0	26.46	26.46
01 January 2025- 23 March 2025	0	6.62	6.62
Total	0	132.30	132.30
Total Number of crediting years	5		
Annual Average over the crediting period	**0**	**26.46**	**26.46**

SDG 7: Affordable and Clean Energy

Year	Baseline estimate	Project estimate	Net benefit
24 March 2020- 31 December 2020	0	12,264	12,264

2021	0	73,319	73,319
2022	0	155,724	155,724
2023	0	247,500	247,500
2024	0	360,000	360,000
01 Jan 2025- 23 March 2025	0	360,000	360,000
Total	0	360,000	360,000
Total Number of crediting years	5		
Annual Average over the crediting period	**0**	**72,000**	**72,000**

SDG 8: Decent Work

Year	Baseline estimate	Project estimate	Net benefit
24 March 2020- 31 December 2020	0	30	30
2021	0	30	30
2022	0	30	30
2023	0	30	30
2024	0	30	30
01 Jan 2025- 23 March 2025	0	30	30
Total	0	30	30
Total Number of crediting years	5		
Annual Average over the crediting period	**0**	**30**	**30**

SDG 15: Life on land

Year	Baseline estimate ($B_{b,y}$)	Project estimate(B_y	Net benefit
24 March 2020- 31 December 2020	0	15,562	15,562
2021	0	119,995	119,995

2022	0	254,858	254,858
2023	0	405,059	405,059
2024	0	589,176	589,176
01 Jan 2025- 23 March 2025	0	132,363	132,363
Total	0	1,517,013	1,517,013
Total Number of crediting years	5		
Annual Average over the crediting period	**0**	**303,403**	**303,403**

SDG 13: Climate Action

Amount of EX-Ante CO_2e emissions reduced by the project per year is as follows:

Year	Baseline estimate	Project estimate	Net benefit
24 March 2020- 31 December 2020	29,997	-	29,997
2021	231,295	-	231,295
2022	491,251	-	491,251
2023	780,768	-	780,768
2024	1,138,774	-	1,138,774
01 Jan 2025- 23 March 2025	255,135	-	255,135
Total	2,927,221	-	2,927,221
Total Number of crediting years	5		
Annual Average over the crediting period	**585,444**	**-**	**585,444**

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 1

Data / Parameter	By
Unit	Fraction

Description	Quantity of fire wood that is saved in the year y (tonnes per ICS in year y)
Source of data	KPT report
Value(s) applied	2.0
Measurement methods and procedures	The baseline and project KPTs will be determined using either paired / independent or single sample tests. 90/30 rule for sample size determination will be applied in case of paired or independent sampling. 90/10 rule will be applied in case of single sampling. In case desired precision is not achieved, lower bound value of the 90% confidence interval should be applied.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	Transparent data analysis and reporting. KPT have been done for paired sample, and the 90/30 rule was respected as demonstrated in the KPT report and the ex-ante calculation sheet.
Purpose of data	Reporting on SDG 1
Additional comment	-

SDG 7

Data / Parameter	ACS
Unit	Number of units
Description	Number of households having access to affordable, reliable and modern project ICS.
Source of data	Project database
Value(s) applied	400,000 during the crediting period
Measurement methods and procedures	The total number of ICS sold/distributed is summed up in the database
Monitoring frequency	continously
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Reporting on SDG 7
Additional comment	-

SDG 8

Data / Parameter	NDJ
Unit	Number of employees earning above minimum wages
Description	Number of decent jobs that have been created
Source of data	Project developer Human resources management records
Value(s) applied	30 (ex-ante)
Measurement methods and procedures	OBEN E.A.C Human resources management data will be check to determine how many people are hired and work on this project.
Monitoring frequency	Annually or at each monitoring period
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Reporting on SDG 8
Additional comment	-

SDG 15

Data / Parameter	WS
Unit	Tons of wood
Description	Quantity of wood saved by the project
Source of data	Project database
Value(s) applied	303,403
Measurement methods and procedures	The number of cookstoves operational during a period will be multiplied by the results of KPT (2t of wood saved by ICS per year)
Monitoring frequency	Calculated at each monitoring period
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Reporting on SDG 15
Additional comment	--

SDG 13

Data / Parameter	fNRB,i,y
Unit	Fractional non-renewability

Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	Calculated in line with CDM Tool 30 EB 108 Annex 11 v3.0 2020
Value(s) applied	0.8183
Measurement methods and procedures	CDM Tool 30 EB 108 Annex 11 v3.0 2020
Monitoring frequency	Fixed for this crediting period. However, this parameter will be monitored before next renewal period
QA/QC procedures	--
Purpose of data	Calculation of emission reduction
Additional comment	--

Data / Parameter	Pb,y
Unit	tonnes/household-year
Description	Quantity of fuel that is consumed in baseline scenario b during year y
Source of data	Field performance Test
Value(s) applied	4.04
Measurement methods and procedures	Kitchen Performance Test
Monitoring frequency	Updated every two years (after the kitchen performance test)
QA/QC procedures	The equipment used for testing, is externally calibrated or at the time of use so measurements are done with the necessary guarantees. The measurements are done by an independant laboratory.
Purpose of data	Calculation of emission reduction
Additional comment	KPT results showed a firewood consumption of 11.1 kg of firewood per households and per dyt hus 4,051 kg per year

Data / Parameter	Pp,y
Unit	tonnes/household-year

Description	Quantity of fuel that is consumed in project scenario b during year y
Source of data	Field performance Test
Value(s) applied	2.0
Measurement methods and procedures	Kitchen performance Test
Monitoring frequency	Updated every two years
QA/QC procedures	The equipment used for testing, is externally calibrated or at the time of use so measurements are done with the necessary guarantees. The measurements are done by an independant laboratory.
Purpose of data	Calculation of emission reduction
Additional comment	KTP results showed a consumption of 5.5 kg after introduction of ICS, thus 2.007 kg/year. These results are found conservative as the test were conducted on mobile ICS and used for as value for each type of ICS. Mobile ICS have lower efficiency than fixed ICS.

Data / Parameter	Up,y
Unit	Fraction
Description	Usage rate in project scenario p during year y
Source of data	Annual usage survey
Value(s) applied	0.9
Measurement methods and procedures	Sampling surveys (telephonic / physical) may be conducted to record the continued operation of project devices. The usage rate shall be calculated for each age (simple random / stratified random sampling to be applied as applicable)
Monitoring frequency	Annual or more frequently, in all cases on time for any request for issuance
QA/QC procedures	OBEN E.A.C will provide guidance and training to enumerators for conducting surveys to meet specific requirement of the methodology, if any. The value obtained will be tested to determine if the desired precision was met.
Purpose of data	Calculation of emission reduction
Additional comment	Althought surveys will be done, the usage rate will be capped at 0,9 in accordance with TPDDTEC Cookstoves usage rate guidelines good practices

Data / Parameter	Np,y
Unit	Project technologies-days credited (units)
Description	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y
Source of data	Technologies distributed in the project database for project scenario p through year y
Value(s) applied	21,900,000
Measurement methods and procedures	The number of cookstoves distributed is multiplied by the number of days in the monitoring period.
Monitoring frequency	Continuous
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions
Additional comment	

Data / Parameter	LEp,y
Unit	t_CO2e per year
Description	Leakage in project scenario p during year y
Source of data	Leakage assessment
Value(s) applied	0
Measurement methods and procedures	Qualitative / quantitative assessment
Monitoring frequency	Every two years
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	For leakage emissions
Additional comment	

B.7.2. Sampling plan

The sampling approach is to be used for following surveys and tests as summarized below:

Ongoing monitoring:
Usage survey: An annual usage survey determines the drop off rates as project technologies age and users switch back to the baseline technology. The usage parameter will be weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario. The minimum total sample size is 100 randomly selected households, with at least 30 samples for project technologies of each age being credited. The majority of interviews will be conducted in person by OBEN E.A.C staff or by hired externals which would be trained before and include expert observation by the interviewer within the kitchen in question, while the remainder may be conducted via telephone by the same interviewers on condition that in-kitchen observational interviews are first concluded and analyzed such that typical circumstances are well understood by the telephone interviewers. The GS cookstove usage rate guidelines will be followed.

Monitoring survey: Along with the usage survey, a monitoring survey is carried out annually to assess end-user characteristics such as technology use, fuel consumption and seasonal variation.

Leakage Assessment
Leakage assessment regarding the potential leakage sources will be completed every other year together with the Usage Survey and therefore no separate sampling approach is used for the leakage assessment.

At least with every 2nd monitoring survey a leakage assessment will be conducted. The leakage assessment evaluates if the project has in any way lead to an increase in emissions outside of the project boundary and if the increase will be directly attributed to the project activity. The leakage assessment is done in the following:

Potential leakage source	**Probability**	**Justification**
The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.	---	--
Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.	--	--
The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.	--	--

The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.	--	--
By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.	--	--

The PP will update the project fuel consumption by carrying out biennial *project KPTs* to account for changes in the project scenario over time as project technologies age. Alternatively, the PP will monitor following Annex 8 of TPDDTEC the degradation in the performance of cookstove efficiency following the WBT and accordingly adjust the project fuel consumption level.

In case of choosing the first option, KPTs would follow the 90/30 rule in case of a paired or independent sampling or the 90/10 rule in case of single sample tests. In case that the 90/30 rule (in case of paired or independent sampling) or 90/10 rule (in case of single sample test) is not met, additional random samples will be taken or the upper bound of the 90% confidence interval will be applied. The procedures as outlined in section 7 and Annex 4 of the applied methodology will be followed.

In case PP opts for the *ageing test approach* instead of biennial project KPTs, annual WBTs would be conducted on a representative sample of each age group. The sample size of each age group will be big enough so that the results comply with the 90/10 rule. The WBT shall be carried out along with the project KPTs prior to 1st issuance and then subsequently WBTs shall be carried out annually to monitor the degradation in the efficiency of the ICS. The WBTs should be conducted in the last 3 months of the monitoring period or after the monitoring period, provided it is representative of annual conditions. Choosing the ageing test approach, the PP would have to ensure to

a) raise additional questions in the monitoring survey related to the frequency of usage of both the project and baseline devices or capture the number of meals cooked or

b) carry out measurement campaigns using data loggers such as stove utilization monitors (SUMs) to take into account for the parallel use of baseline stoves in the project scenario or

c) capture the number of meals cooked on both project and baseline devices. The measurement campaign would comprise of at least 100 randomly selected households for at least 90 days, with at least 30 samples for project technologies of each age being credited.

The sampling frame for the aforementioned ICS surveys comprises of all households making part of the database.

The sampling frame for the KPTs/WBTs consist of all households using the project ICS.

The surveys/tests will be carried out by OBEN E.A.C staff or by hired externals which would be trained before. Quality of the data will be checked by the carbon consultant.

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B.7.3. Other elements of monitoring plan

OBEN E.A.C keeps records of all distributed ICS in an electronic database. As a minimum, following information will be recorded in the database:

- Unique identification number of the ICS
- Date of distribution
- Name of the beneficiary
- Geographic area (Province/commune) of the beneficiary
- Model type of the ICS

The unique identification number of each ICS entered into the distribution database will be linked to a distribution date (recorded during distribution). Thus, for any monitoring period it is possible to calculate the period of time for which the stoves included in the emissions reduction calculations are deemed operating. The sum of the operating fractions of all appliances determines the equivalent full-time appliances for the monitoring period. Furthermore, separate database will be kept for each model of ICS sold.

All data will be stored for at least two (2) years after the expiry of the crediting period.

OBEN E.A.C is responsible for overall monitoring organization. The data collection & consolidation and results analysis are implemented by an adequately trained monitoring team, well aware of Gold Standard requirements and supervised by OBEN E.A.C. The monitoring team consists of one monitoring manager and one data manager.

Role	Responsibilities	Identity
Monitoring manager	- In general, ensure that the project follows the monitoring plan - Ensure that the equipment and measurements are in line with the measurement methods, recording frequency and archiving approaches in monitoring plan - Ensure that monitoring data collected is consolidated and entered in electronic database - Ensure that monitoring team receives proper training - Ensure a coherent and standard monitoring report for each CPA	Mr. Claver NDIZEYE
Data manager	- Collect monitoring data - Enter data in electronic database and archive hardcopies[10]	Mr. Christian Bob NYABUZANA

[10] *Data monitored and required for verification and issuance are kept and archived for at least two years after the end of the final crediting period or the last issuance of GSVERs, whichever occurs later.*

Consultant	- Carry out sample size determination and emission reduction calculations	AERA Group

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

Project start date is 24 March 2020 (the date of installation of first unit under the Project activity).

C.1.2. Expected operational lifetime of project

Project Life operational lifetime will be 15 years (5 years for each Design Certification Renewal Cycle and in total two Design Certification Renewal Cycles)

C.2. Crediting period of project

C.2.1. Start date of crediting period

24 March 2020 (the start date of project operation) or two years prior to the date of Project Design Certification, whichever is later

C.2.2. Total length of crediting period

15 years (5 years for each Design Certification Renewal Cycle and in total two Design Certification Renewal Cycles)

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in Appendix 1, ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3.1 Community Health, safety & working conditions	The amount of woodfuel consumed will be measured in the determination of the emission reductions
Principle 6.1 1)Employment in legal compliance	Ensure all production personnel have legally compliant employment contracts
Principle 6.1 3)Employment contracts documented	Ensure all production personnel have documented employment contracts
Principle 6.1 5) Appropriate equipment, Training of workers	Assess training material and competence of manufacturers and end users.
Principle 9.4 Release of pollutants	The Emission Reduction (ER) calculations of the project – positive benefit

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project aims to promote the use of women in manufacture of the ICS. This is important in a region where women have experienced negative discrimination traditionally and do not have significant employment opportunities. The project aims to promote the use of women in manufacture of the ICS. There will be no discrimination in the project and there will be equal pay for equal work in the manufacturing of the ICS.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Government of Burundi's efforts to promote gender equality focus largely on implementing the Beijing Declaration and the Beijing Platform that U.N. member countries adopted in 1995 end[11]. The project does not have any component that is at odds with this guidance.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, the project will be ensuring that women will increase their access to, or control of benefits yielded buy the project, specifically around fuel collection and cooking, and hence an expert is not required for this.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, as the project activity is increasing the role of women through it's application, and women have formed a major part of the local stakeholder consultation, and the UN does not discriminate on gender for returnees receiving aid packages (and hence stoves), an expert would not be required.

[11] US AID – United States Agency for International Development: "USAID/Burundi Gender analysis. Final Report 2017", https://banyanglobal.com/wp-content/uploads/2017/07/USAID-Burundi-Gender-Analysis-Final-Report-2017.pdf, accessed 19 January 2021

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

A first physical local consultation meeting has been conducted in Kinyinya located in the Ruyigi province (East of Burundi) on 20th August of 2021. Several means (phone, mails, posters were used to invite people.

A second hybrid (physical and remote for people out of Burundi) consultation have been done on 20th October 2021. Participants from Burundi met at OBEN E.A.C offices at Bujumbura, while remote participants used Microsoft Team to attend to the meeting.

Individuals, who couldn't attend the local stakeholder consultation meeting, were able to comment the non-technical summary of the program via mail, email or telephone. The stakeholders who didn't reply to the invitation were reminded on the meeting via telephone.

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E.1. Summary of stakeholder mitigation measures

>>For this project, 2 LSC were held, one physical on 20/08/2021, and one hybrid (physical and remote) on 20/10/2021. All the comments and suggestions provided by the stakeholders participating in the 2 LSC meeting were in form of clarifications and advice. No mitigation measures are triggered to be included in the project design as a result from the local stakeholder consultation meeting as all the comments were either clarified by the project developer or they were already considered in the project design. Following the last consultation held on 20/10/2021, the SFR was conducted for 2 months. No comments/complaints were received during the SFR period. Please see the Stakeholder Consultation Report for more details

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	At OBEN offices in BUJUMBURA: Burundi : B.P.4942 Bujumbura (Burundi) At UNHCR Offices: 78, Avenue du Large, Kinindo, Bujumbura (Burundi)
GS Contact (mandatory)	help@goldstandard.org

	Telephone AERA Group : +33 6 42 96 09 78
	e-mail AERA Group: info@aera-group.fr
Other	Telephone OBEN: +257 76 60 26 00
	E-mail OBEN : loicqueen@yahoo.fr

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into SECTION D above. Please refer to the instructions in the Guide to Completing this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No	The project will be within the borders of the Republic of Burundi. The Republic of Burundi is a member of the United Nations and the African Union. It has ratified many UN Human Rights Conventions and thus has made binding international commitments to adhere to the standards laid down in these universal human rights documents. Hence, the project does violate human rights obligations adopted by the host country.	Not required
2. The Project shall not discriminate with regards to participation and inclusion	No	The project is of a non-discriminatory nature regarding ethnicity, religion, gender and age.	Not required

Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women	No	The project aims to promote the use of women in manufacture of the ICS. This is important in a region where women have experienced negative discrimination traditionally and do not have significant employment opportunities.	Not required
2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work	No	The project aims to promote the use of women in manufacture of the ICS. There will be no discrimination in the project and there will be equal pay for equal work in the manufacturing of the ICS.	Not required
3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks	No	The Government of Burundi's efforts to promote gender equality focus largely on implementing the Beijing Declaration and the Beijing Platform that U.N. member countries adopted in 1995 end[i]. The project does not have any component that is at odds with this guidance.	Not required

4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	Not required	Not required	Not required
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	Yes (positive impact)	The project activity reduces the amount of wood fuel that is combusted (often indoors) for cooking. This has a resultant positive benefit in as much as a reduction in potentially harmful emissions indoors (containing particulates and higher non breathable gases. According to the World Health Organisation (WHO)[ii], Indoor air pollution is around 100 times more harmful when cooking on indoor inefficient stoves, resulting in close to 4 million deaths per annum globally[iii].	Measurement of amount of wood consumed
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project activity relates to the dissemination and use of domestic ICS devices in private homes. Traditional cooking fires are not considered cultural significant.	Not required

>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)? >>	No	The project activity relates to the dissemination and use of domestic ICS devices in private homes. No relocation of people is required. Contrarily, people being supplied with the ICS in the project activity are actually returnees from previous displacements and the project is contributing to their sustainable return.	Not required
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? >>	No	The project activity relates to the dissemination and use of domestic ICS devices in private homes. No land use is required. The manufacturing site has the necessary permits for manufacture.	Not required
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project is not operating in area claimed by indigenous people.	Not required

>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	While the project activity involves the dissemination, at no cost to the individuals, of ICS devices via aid packages to returnees; there will be contracts and transparent governance of the contracts with aid organisations who will be packing, transporting, distributing and recording the dissemination of the ICS to the returnees.	Not required
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions	Potentially	A number of individuals will be employed in the manufacture of the ICS devices. All personnel will be employed legally in terms of Burundi labour legislation, which meets or exceeds the ILO requirements.	Ensure all individuals employed directly by the project activity have legal labour contracts.
2. Workers shall be able to establish and join labour organisations	No	Burundi is a member of the ILO since 1963. They have national legislation covering the right of workers to form collective representation[iv] - all personnel will	Not required

3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families'	Potentially	be employed legally in terms of Burundi labour legislation. A number of individuals will be employed in the manufacture of the ICS devices. All personnel will be employed legally in terms of Burundi labour legislation, which meets or exceeds the ILO requirements.	Ensure all individuals employed directly by the project activity have legal labour contracts.

property requires an Expert Stakeholder opinion)			
	Yes	The minimum age for employment in Burundi is 16[v], all personnel will be employed legally in terms of Burundi labour legislation, which meets or exceeds the ILO requirements. No children will be involved in the project activity, except as final users in a family environment.	Ensure all individuals employed directly by the project activity have legal labour contracts.
5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures	Yes	Training and correct manufacture and use of the ICS is important for the effective implementation of the project activity. All personnel involved in the manufacture of the ICS devices will be trained and competence assured through training, monitoring and quality assurance on product. Training sessions and training material will be made available for the correct and efficient use of the ICS devices by final end users. This will be via material distributed with the ICS devices and extension workshops.	Assess training material and competence of manufacturers and end users.

Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation? >>	No	The project activity will only create positive economic effects for those using the ICS. There is no up-front payment, it will reduce the amount of wood required for cooking and there are associated health benefits mitigating the need for possible medical costs.	Not required
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario? >>	No	The ICS will consume less wood for combustion due to combustion and thermal efficiency of the stoves, hence the resultant CO_2 emissions from the same amount cooking will be significantly lower (as per project emission reduction calculations).	Not required
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users? >>	No	The fuel source is already being utilised for cooking. The project activity allows for the more efficient use of the same wood resource.	Not required

Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project activity is not related to water use or water consumption at all.	Not required
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The project activity is not related to water use or water consumption at all.	Not required
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project activity is not related to land use or crop production at all.	Not required
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding,	No	The project is located within domestic households. There is no activity which can adversely affect the natural	Not required

drought or other extreme climatic conditions? >>		system or lead to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions	
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)? >>	No	The project activity is not related to land use or crop production at all, and hence no exposure to or influence on the use of GMOs	Not required
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment? >>	Yes (positive)	The project activity will release CO_2 from combustion of wood in cooking into the environment. However, the increased efficiency of the ICS devices has a resultant net positive benefit to the environment with less pollution (including indoor particulates in fire smoke) being released into the environment.	The Emission Reduction (ER) calculations of the project

Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials? >>	No	The project activity only utilises wood for combustion and no hazardous chemicals or waste is produced.	Not required
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers? >>	No	The project activity is not related to land use or crop production at all, and hence no pesticides and/or fertilisers will be used.	Not required
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests? >>	No	The project activity is not related to forest harvesting at all, contrarily, wood use will be reduced due to more efficient cooking using the ICS devices.	Not required
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives? >>	No	The project activity is not related any impact on a food source.	Not required

Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry? >>	No	The project activity is not related to animal husbandry in any way	Not required
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified? >>	No	The project activity is not related to forest harvesting at all, particularly HCV. Contrarily, wood use will be reduced due to more efficient cooking using the ICS devices.	Not required
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects? >>	No	The project activity will not impact areas negatively where any endangered species may be present.	Not required

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	AERA Group SAS
Registration number with relevant authority	812.807.329
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APPENDIX 3- LUF ADDITIONAL INFORMATION

>>N/A

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Annex A of Principles and Requirements for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption

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[i] US AID – United States Agency for International Development: "USAID/Burundi Gender analysis. Final Report 2017", https://banyanglobal.com/wp-content/uploads/2017/07/USAID-Burundi-Gender-Analysis-Final-Report-2017.pdf

[ii] "Around 3 billion people still cook and heat their homes using solid fuels (i.e. wood, crop wastes, charcoal, coal and dung) in open fires and leaky stoves. Most are poor, and live in low- and middle-income countries.
Such inefficient cooking fuels and technologies produce high levels of household air pollution with a range of health-damaging pollutants, including small soot particles that penetrate deep into the lungs. In poorly ventilated dwellings, indoor smoke can be 100 times higher than acceptable levels for fine particles. Exposure is particularly high among women and young children, who spend the most time near the domestic hearth." – World Health Organization – Regional Office for Africa: AIR POLLUTION, https://www.afro.who.int/health-topics/air-pollution, accessed 19 January 2021

[iii] World Health Organization: FACT SHEET HOUSEHOLD AIR POLLUTION and HEALTH, https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health, accessed 19 January 2021

[iv] International Labour Organization (ILO): Convention collective interprofessionnelle nationale du Travail conclue le 3 avril 1980 et régie par les dispositions du chapitre premier du titre XI de l'arrêté-loi n° 001/31 du 2 juin 1966 portant Code du travail au Burundi.,

https://www.ilo.org/dyn/natlex/natlex4.detail?p_lang=en&p_isn=34372&p_country=BDI&p_count=237&p_classification=01.02&p_classcount=3, accessed 19 January 2021

[v] Government of Burundi: Article 3 of the Labor Code; Article 3 of the Ministerial Ordinance to Regulate Child Labour